

RESOLUCIÓN DE LAS ACTIVIDADES

1.

a) $f'_x = 16x^3$ $f'_y = 2$

b) $f'_x = 7e^y$ $f'_y = 7xe^y$

c) $f'_x = \frac{1}{y+2}$ $f'_y = \frac{-x+3}{(y+2)^2}$

d) $f'_x = 5 \cdot (y-1) \cdot x^{y-2}$ $f'_y = 5x^{y-1} \cdot \ln x$

e) -----

f) $f'_x = \frac{y^2}{(x^2+y^2)^{3/2}}$ $f'_y = \frac{xy}{\sqrt{x^2+y^2}}$

g) $f'_x = 2xy \cos(xy) - x^2 y^2 \cdot \text{sen}(xy)$ $f'_y = x^2 \cdot (\cos(xy) - xy \text{sen}(xy))$

h) $f'_x = \left[y \ln(2x^3 + 1) + \frac{6x^3 y}{2x^3 + 1} \right] \cdot (2x^3 + 1)^{xy}$ $f'_y = x \cdot (2x^3 + 1)^{xy} \cdot \ln(2x^3 + 1)$

2. Ver ejercicio desarrollado en la infografía.

3. Se verifica.

4.

a) $F_{x,x} = e^x \cdot \ln(3 - y^2)$ $F_{x,y} = \frac{-2ye^x}{3-y^2}$ Son iguales.
 $F_{x,y} = e^x \cdot \frac{-2y}{3-y^2}$ $F_{y,x} = \frac{-2y}{3-y^2} \cdot e^x$

b) $F_{x,x} = 2x + y \cdot \cos x$ $F_{x,y} = x^2 - \sin y + \sin x$ Son iguales.
 $F_{x,y} = 2x + \cos x$ $F_{y,x} = 2x + \cos x$

c) $F_{x,x} = \frac{1}{x^2}$ $F_{x,y} = \frac{-1}{y^2}$ $F_{x,y} = F_{y,x} = 0$

d) $F_{x,x} = \frac{1}{x-y}$ $F_{x,y} = -\frac{1}{x-y}$ Son iguales.
 $F_{x,y} = \frac{1}{(x-y)^2}$ $F_{y,x} = \frac{1}{(x-y)^2}$

e) -----

f) $F_{x,x} = \frac{-x}{\sqrt{9-x^2-y^2}}$ $F_{x,y} = \frac{-y}{\sqrt{9-x^2-y^2}}$ Son iguales.
 $F_{x,y} = \frac{xy}{\sqrt{(9-x^2-y^2)^3}}$ $F_{y,x} = \frac{xy}{\sqrt{(9-x^2-y^2)^3}}$

5. Se verifica.